Differential Causal Impacts of Emigration Rates on Male & Female Schooling Decisions

Shravan Hari, Ben Miele, and Sahil Najeeb

February 18, 2026

Abstract

We utilize publicly available data from the Mesoamerican Migration Project to analyze the causal effects of emigration rates in Mexican communities on local household schooling decisions using a two-stage least squares regression with instrumental variables. Our analysis reveals that emigration rates within communities decrease continuation rates at the margin from primary to middle school for male students after controlling for factors like wealth and community-level remittances, whereas the impact is much more ambiguous and insignificant for female students. We therefore suggest that the decrease occurs by way of a substitution effect, where an increase in the perceived returns to emigration makes education for marginal students relatively less attractive.

1 Introduction

The idea that migration and education in developing economies are connected is intuitive. An individual who emigrates from a poorer nation to a wealthier one dramatically increases their earning capacity, and is able to remit funds to finance the education of family members, particularly children, in their home country. This mechanism of emigration focuses on the **income effect** that emigration to a wealthier nation can provide. However, emigration can also be understood through an alternative mechanism by means of the following example.

Consider a Mexican household with a school-aged son noticing increasing levels of emigration to the United States within their community, in particular among individuals with low incomes and low levels of education. Observing this trend, the household might perceive an increase in the likelihood that their son will emigrate to the U.S. as an adult, where he would likely engage in low-skilled, manual (but remunerative) work in sectors like agriculture, manufacturing, and construction.[1] All else equal, this would cause them to substitute away from investment in their son's education, as the "outside option" of emigration is now perceived to be more valuable.[2] When this is understood as a mechanism by which emigration affects education decisions, the emphasis is laid on the negative **substitution effect** caused by emigration. To isolate this effect, one needs to account for the aggregate and individual income effects of emigration rates, which can be achieved through controls on remittances.

This paper focuses on this mechanism with particular regard to Mexican families' decisions to enroll their children in *secundaria* school (analogous to middle school in the United States) following the completion of primary school. That is, we attempt to analyze how emigration rates within these communities affect this decision, and in particular how this effect is differently pronounced for male and female children. Males are perceived as having higher returns to emigration given that the jobs occupied by Hispanic immigrants in the US often center around manual labor, which would make the substitution effect more likely to dominate

[1] Federal Reserve Bank of St. Louis, *Unemployment Rate - Hispanic or Latino.* `https://fred.stlouisfed.org/series/LNS14000009`

[2] This rests on the assumption that what drives the increase in emigration is a higher perceived return to emigration.

the income effect for males than for females.[3] Additionally, we may see females as having higher returns on education than males,[4] creating a further gendered differential between the substitution effects.

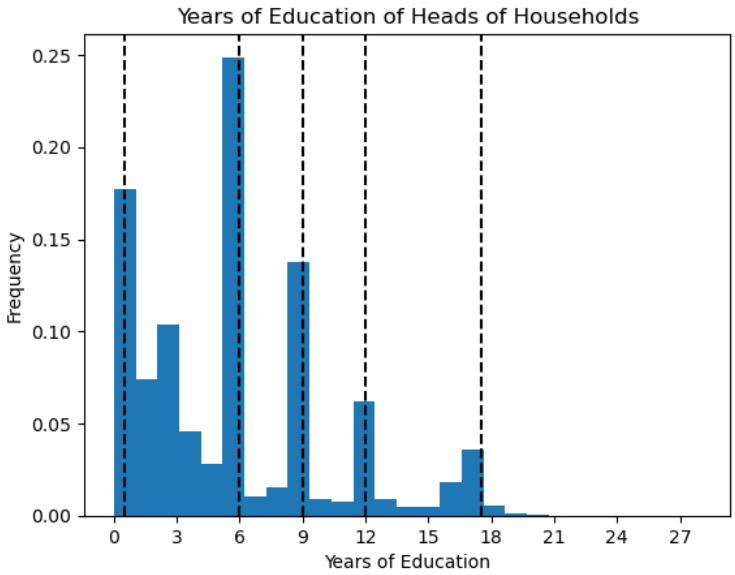

Figure 1: Years of Education of Household Heads in Mexico

The above graph is a histogram of the number of years of education of household heads from our dataset (n = 27,410). The dotted lines indicate stages of transition in Mexico's educational system: 6 years in primary school (*primaria*), 3 years in secondary school (*secundaria*), 4 years in high school (*bachillerato*), and university (*licenciatura*).[5] The mode of the graph is at 6, with a sharp cutoff the year after, indicating that the sharpest drop in years of education occurs between years 6 and 7: precisely the cutoff between *primaria* and *secundaria*.

To estimate the effect of emigration rates on parents' enrollment decisions, we assemble observations of 6,953 Mexican individuals born between 1963 and 2004 to parents of non-emigrants who do report receiving remittances from the United States. Every individual in our sample either finished school immediately after primary school, or began secondary school, but dropped out after one year. Essentially, our design allows us to answer this question: Among students on the margin between primary and secondary school, do emigration rates affect their likelihood of stopping at primary versus continuing slightly beyond? Utilizing a shift-share design which exploits differential community-level exposure to U.S. labor shocks, we find that a 1 percentage point increase in the rate at which individuals aged 15-65 in a community begin a migration spell over the course of a given child's primary education (ages 5-12) has a weak and insignificant negative effect on the likelihood that the child begins secondary school and quits after one year. Segmenting our sample by sex, we find that emigration rates *do* significantly decrease the likelihood that boys begin secondary school and quit after one year, with no significant effect for girls. This effect is robust to controls for household wealth, parental educational attainment, local remittance rates, migration prevalence, and household emigration experience. This result aligns with the intuition laid out above; as male emigrants have higher earning capacity, the returns to emigration for men are higher, so we would expect the substitution effect to be more pronounced.

[3] Leibbrand, C. *Unequal Opportunity? Racial, Ethnic, and Gender Disparities in the Returns to Internal U.S. Migration.* `https://pmc.ncbi.nlm.nih.gov/articles/PMC7286628/`

[4] Dougherty, C. *Why are the results to schooling higher for men than for women?.* Journal of Human Resources, October 2005. `https://jhr.uwpress.org/content/40/4/969`

[5] "E2 -Grade Levels and Structure of the Educational System of Mexico Grade Levels and Structure of the Educational System of Mexico." n.d. https://www.cde.state.co.us/migrant/gradelevelsandstructureoftheeducationalsystemofmexico.

2 Data

We utilize data that was predominantly sourced from the *Mesoamerican Migration Project* (MMP) organized by Brown University. The MMP174 database is a rich source of publicly available cross-sectional information containing almost 180,000 observations across households in 174 communities in Mexico, cataloging their education and migration histories. The PERS file provides general demographic information (including age, sex, education levels, etc.) and a number of migration measures for each member of a surveyed household (including whether they made any trips to the US, the duration of the trips, etc.); we obtained data on remittances from the MIG file. To construct our instrument, we used prevalence ratios from the years 1950-70 (see Subsection 2.3) from the PRATIO file and data on the Hispanic unemployment rate, sourced from FRED.

In addition to the individual-level variables described above, we construct the remaining objects used in the analysis as follows. Our endogenous regressor, the community emigration rate $E_{j,t}$, is measured using the life-history module by identifying new migration "spell starts" to the United States among working-age individuals (ages 15–65) in each community-year and aggregating these flows with survey weights and expressing the resulting flow as percentage points of the community working-age population. For each child we then average this community emigration over the child's primary-school years (ages 5–12), so that each observation is assigned a single exposure measure reflecting the migration environment during the period in which the schooling decision was made. To separate the substitution channel from income effects, we control for local remittance intensity: we use the migration/remittance module to identify migrants who send positive monthly remittances and compute (i) the share of the community sending positive monthly remittances in a given year and (ii) the average monthly remittance amount among senders; we proxy per-capita remittance intensity by the product of these two objects and average this measure over the child's primary-school window.

We also include household-level covariates $X_{i,j,t}$: parental years of schooling (constructed from the household head and spouse), a proxy for household resources during childhood based on a log wealth index formed from land, property, and business ownership observed in the life-history panel during ages 5–12, and an indicator for whether any non-parent household member has ever emigrated, which captures direct household exposure to migration networks beyond the parents. Finally, our instrument $Z_{j,t}$ is a Bartik-style shift-share interaction between a community's pre-period migration network strength, measured by the prevalence ratio averaged over 1950-1970, and a common U.S. labor-market shock given by the annual U.S. Hispanic unemployment rate; we average the shock over each child's primary-school years to match the construction of exposure. All regressions include community fixed effects and cohort (year-of-birth) fixed effects, and we report standard errors clustered at the community level.

2.1 Sample Definition

We made a few restrictions on the data obtained from the MMP174 database to subset a sample that would provide meaningful results and interpretation:

- Only children that had attended and completed the six years of primary school were chosen to be part of the sample. Consequently, we only chose individuals over the ages of 14 at the time of the survey since they are the only ones that could have completed their seventh year of education.
- Only children that did *not* complete 8 or more years of education were chosen. This restricts to students on the margin between primary and secondary school, allowing us to examine the treatment effect on those who are plausibly deciding whether to continue schooling.
- We excluded any households in which (a) the head of the household was missing, (b) either the head of the household or their spouse had ever made a trip to the US to restrict attention to the substitution effects in those households with no emigration experience.
- We only included data from after 1974 for reasons discussed in Subsection 2.3.

2.2 Empirical Strategy & Identification

In the following equation t will denote the usual time index, j a 'community' index, and i a household index. Then we write

$$Y_{i,j,t} = \beta E_{j,t} + \delta X_{i,j,t} + \varepsilon_{i,j,t},$$

where in a given year t:

- $Y_{i,j,t}$ is an indicator for whether child i in community j continues beyond primary school (defined in Subsection 2.1),
- $E_{j,t}$ is the community emigration flow rate during the child's primary-school window, measured as new U.S. migration spell starts (percentage points) averaged over ages 5–12,
- $X_{i,j,t}$ is a set of household-level covariates (including community-wide per-capita remittances, parents' years of education, household wealth, whether any non-parent household member has ever emigrated, sex, single-parent status, and migration prevalence during primary school), and
- $\varepsilon_{i,j,t}$ is the random error term.

Then β is the effect of a one percentage point increase in community migration spell starts (averaged over ages 5–12) on the probability that a marginal student continues beyond primary school.

Since the income of several of the households was unobservable we substituted it with the following measure of wealth:

$$W_i = \log(1 + \text{hectares}_i) + \log(1 + \text{land}_i) + \log(1 + \text{property}_i) + \log(1 + \text{business}_i),$$

where hectares_i is the maximum of zero and the number of hectares owned by the individual in person-year; likewise, land_i, property_i, and business_i, represent the number of land parcels, properties, and businesses owned in person-year. This provides a reasonably holistic measure of household income.

2.3 Endogeneity Concerns & Shift-Share IV Design

Community-wide average emigration rates, $E_{j,t}$, present endogeneity concerns for a few reasons. Emigration is certainly not exogenous, as household needs and household-specific demand fluctuations affect an individual's decision to emigrate. In addition, even with community/individual level and year fixed effects, there is still unobserved variation in school and municipal policy that may affect migration decisions. Therefore, the setting is appropriate for an instrumental variable design.

We use a Bartik-style shift-share instrument to account for our endogenous regressor. Consider the following design:

$$Z_{j,t} = P_j \times U_t$$

P_j is the community-specific prevalence ratio, defined as "number of people with international migratory experience divided by the total number of people alive."[6] P_j functions essentially as a proxy for migrant networks, indicating how connected different communities are to migratory networks and the extent of their migratory development. This is a unit that varies over time, but we take a fixed historical prevalence ratio average from 1950 to 1970. This allows P_j to be exogenous to intertemporal emigration after 1970.

Here, U_t is the Hispanic unemployment rate in the United States, taken from FRED data. We only observe Hispanic unemployment data starting in 1974, since it was the first year where Hispanic origin became consistently defined, and time series data of unemployment rates become suitable for analysis. As a result we also only consider observations where at least some primary schooling occurred after 1974, in order to be plausibly exogenous from both components of $Z_{j,t}$. With these components, we have time variation from U_t and spatial variation from P_j, allowing us to run a 2SLS to recover exogenous variation in our endogenous regressor, $E_{j,t}$.

[6] Mexican Migration Project. n.d. *PRATIO Codebook* (MMP Codebook No. MMP174). Accessed February 18, 2026. `https://mmp.research.brown.edu/images/pratio-codebook-mmp174-.pdf`.

Justification of Instrument

- **Relevance:** Our instrument $Z_{j,t}$ is correlated with the endogenous regressor $E_{j,t}$ as justified in the regression tables below. Intuitively, communities with a higher emigration rate tend to have a higher PRATIO, which provides a baseline level for which time variation is picked up by the unemployment rate U_t, making $Z_{j,t}$ correlated with $E_{j,t}$.
- **Validity:** Using the shift-share design, we note that U_t is caused by exogenous unemployment shocks in the US, and is therefore not connected to unobserved determinants of $Y_{i,j,t}$; the same is true of the prevalence ratio, which is a historical region-specific constant in time. Therefore, $U_t \times P_j$ is not correlated with $\varepsilon_{i,j,t}$.
- **Exclusion:** There is no reason to believe that either of the terms that comprise $Z_{j,t}$ move $Y_{i,j,t}$ except through the regressor $E_{j,t}$. A historical parameter interacted with national shocks is plausibly exogenous to individual schooling decisions, and can only possibly affect it through the mechanism of emigration rates.

3 Results

We run two separate 2SLS regressions, subsetting the male and female populations respectively. The results of the first stage regression are quite strong across sexes, with a p-value of less than $2.2 \cdot 10^{-16}$ in both cases; this is justification that our instrument was highly relevant. The naive OLS estimate without instrumenting is also highly insignificant, reflective of the concerns of endogeneity and reverse causality that might occur if we had not used an instrument for emigration rates.

The results of the 2SLS regression on the male population yielded a coefficient of $\boxed{\hat{\beta} = -0.1131}$ on the emigration rate (and consequently, an estimate of -0.1131 of the Local Average Treatment Effect across time), the interpretation of which is as follows: a 1 percentage point increase in the emigration rate within a community drops the likelihood that a male child in that community continues from primary school to middle school by 11 percentage points. The p-value of this estimate is 0.0479, indicating statistical significance at the 5% level.

The regression on the female population yielded a coefficient of $\boxed{\hat{\beta} = 0.0063}$, a much more ambiguous estimate of the Local Average Treatment Effect across time. The p-value was also > 0.8, indicating no statistical significance whatsoever. Therefore, we compute an at best extremely weak causal effect of emigration rates on female schooling decisions.

Our empirical results strongly corroborate the hypothesis that after controlling for factors like remittances and wealth, community-wide emigration causes families to substitute away from education. A reasonable interpretation of this is that emigration is a more lucrative prospect for males than females and education a more lucrative prospect for females than males, making the (negative) causal effect of emigration on education levels much more pronounced and significant for male children than for female children.

All regression tables can be found on the following pages: Table 1 shows the results of the naive OLS and the non-gender-differentiated IV regressions. Table 2 shows the results of the first-stage regression. Table 3 shows the gender-differentiated IV regression tables and table 4 shows the gender-differentiated first stage results.

Table 1: Migration Exposure and Schooling: OLS & IV Estimates

	Educ = 7 (0/1)	
	(1) OLS	(2) IV
Variables		
Spell starts (pp, primary)	0.000474	-0.034416
	(0.007396)	(0.034693)
Remittances per person (primary)	0.000021	0.000023$^{\bullet}$
	(0.000014)	(0.000014)
Wealth index (primary)	0.002678	0.002800
	(0.003745)	(0.003758)
Mother education (years)	0.007834***	0.007882***
	(0.001975)	(0.001974)
Father education (years)	0.012661***	0.012756***
	(0.002116)	(0.002125)
Other HH ever emigrated	-0.014475	-0.012759
	(0.010080)	(0.010247)
Migration prevalence (pp, primary)	0.005011	0.021422
	(0.003830)	(0.017705)
$\mathbb{I}$(Female)	-0.016756*	-0.017517*
	(0.006959)	(0.006946)
Fixed-effects		
Community FE	Yes	Yes
Cohort FE (YOB)	Yes	Yes
Observations	6,953	6,953
F-stat (1st stage)		499.22

Clustered (community) standard errors in parentheses.
IV instrument: $Z_{j,t}$.
Signif. codes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$, $^{\bullet}p < 0.1$.

Table 2: First Stage: Effect of Instrument on Spell Starts

	Spell starts (pp, primary)
	(1) First stage
Instrument	
$Z_{j,t}$ (primary school exposure)	2.592733***
	(0.662424)
Controls	
Wealth index (primary)	0.001027
	(0.010081)
Remittances per person (primary)	0.000078
	(0.000055)
Mother education (years)	0.001107
	(0.002713)
Father education (years)	0.002903
	(0.003595)
Other HH ever emigrated	0.035797$^{\bullet}$
	(0.020835)
Migration prevalence (pp, primary)	0.471657***
	(0.080674)
$\mathbb{I}$(Female)	-0.022840$^{\bullet}$
	(0.013739)
Fixed-effects	
Community FE	Yes
Cohort FE (YOB)	Yes
Observations	6,953
F-stat (excluded instrument)	499.22

Clustered (community) standard errors in parentheses.
Signif. codes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$, $^{\bullet}p < 0.1$.

Table 3: IV by Sex: Effect of Spell Starts on Schooling

	Educ = 7 (0/1)	
	(1) Boys	(2) Girls
Variables		
Spell starts (pp, primary) [IV]	-0.113170*	0.006322
	(0.056801)	(0.033287)
Wealth index (primary)	-0.001782	0.006068
	(0.005450)	(0.005202)
Remittances per person (primary)	0.000013	0.000031$^{\bullet}$
	(0.000014)	(0.000017)
Mother education (years)	0.006653*	0.007686**
	(0.002724)	(0.002773)
Father education (years)	0.013290***	0.012338***
	(0.002906)	(0.002621)
Other HH ever emigrated	-0.018915	-0.003404
	(0.013141)	(0.013152)
Migration prevalence (pp, primary)	0.059419$^{\bullet}$	0.006512
	(0.032736)	(0.015324)
Fixed-effects		
Community FE	Yes	Yes
Cohort FE (YOB)	Yes	Yes
Observations	3,516	3,431
F-stat (1st stage)	210.89	291.07
Wu–Hausman p-value	0.0208	0.9919

Instrument: $Z_{j,t}$.
Clustered (community) standard errors in parentheses.
Signif. codes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$, $^{\bullet}p < 0.1$.

Table 4: First Stages by Sex: Effect of Instrument on Spell Starts

	Spell starts (pp, primary)	
	(1) Boys	(2) Girls
Excluded instrument		
$Z_{j,t}$ (primary)	2.154876***	3.038480***
	(0.601480)	(0.765349)
Controls		
Wealth index (primary)	-0.010069	0.021257
	(0.012611)	(0.013172)
Remittances per person (primary)	0.000027	0.000126*
	(0.000050)	(0.000062)
Mother education (years)	0.001898	0.000354
	(0.003309)	(0.004032)
Father education (years)	-0.003273	0.008352•
	(0.004094)	(0.004688)
Other HH ever emigrated	0.052244*	0.028437
	(0.023402)	(0.026550)
Migration prevalence (pp, primary)	0.514857***	0.437003***
	(0.066957)	(0.094051)
Fixed-effects		
Community FE	Yes	Yes
Cohort FE (YOB)	Yes	Yes
Observations	3,516	3,431
F-stat (excluded instrument)	210.89	291.07

Clustered (community) standard errors in parentheses.
Signif. codes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$, $^{\bullet}p < 0.1$.

4 References

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